1.4 Solving Multi-Step Equations

Name _____ Period ____ Date _____

Algebra I 1.4 · Student sheet

I can…

- solve two-step equations.
- solve equations using the Distributive Property and by combining like terms.
- write and solve multi-step equations for real-life problems.

Vocabulary · fill in the missing word

Like terms — terms with the same _____ raised to the same power

Distributive Property — a(b + c) = _____

Two-step equation — undo addition or subtraction first, then _____ or division

Part 1 · Error analysis

Each solution has one mistake. Circle it, explain it, and solve correctly.

Student work	The mistake	Correct solution
$2x + 6 = 14 \rightarrow 2x = 20 \rightarrow x = 10$		
$3(x - 4) = 9 \rightarrow 3x - 4 = 9 \rightarrow 3x = 13 \rightarrow x = 13/3$		
$5x - 2x + 4 = 19 \rightarrow 7x + 4 = 19 \rightarrow 7x = 15 \rightarrow x = 15/7$		

Part 2 · Notes: fill in the blanks

Steps: _____ and combine like terms, then undo _____ or subtraction, then multiplication or _____.

Distribute to _____ term inside the parentheses.

Check by _____ the solution into the original equation.

Part 3 · Examples (work them with the class)

Example 1

Solve $-4x + 7 = -13$.

Example 2

Solve $2(3x - 1) + x = 19$.

Example 3

Twice a number plus 8 is 30. Find the number.

Example 4

The angles of a quadrilateral measure x°, 2x°, 90°, and 120°. The angles of a quadrilateral add to 360°. Find x.

Part 4 · Partner practice

Partner 1

Solve $\frac{x}{3} - 4 = 2$.

Partner 2

Solve $-3(x + 2) = 15$.